THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Starting from 21, the hailstone (Collatz) sequence reaches 1 in how many steps? 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 9

2. The next palindrome after 4883 is: 1

(a) 4884 **(b)** 4994 **(c)** 4774 **(d)** 4894

3. The sum of the first 45 counting numbers $1 + 2 + 3 + \cdots + 45$ is: 1

(a) 2070 **(b)** 1080 **(c)** 1035 **(d)** 2025

4. A Kaprekar step on 9821 gives: 1

(a) 8523 **(b)** 8532 **(c)** 8622 **(d)** 8541

5. The greatest 4-digit number whose digit sum is 15 is: 1

(a) 9510 **(b)** 8700 **(c)** 9600 **(d)** 6900

6. The smallest EVEN number that can be formed by using each of the digits 9, 8, 1, 3 once is: 1

(a) 1938 **(b)** 1398 **(c)** 3198 **(d)** 9318

7. The greatest 5-digit palindrome with first digit 6 is: 1

(a) 99999 **(b)** 6996 **(c)** 69896 **(d)** 69996

8. Starting from 24, the hailstone (Collatz) sequence reaches 1 in how many steps? 1

(a) 12 **(b)** 11 **(c)** 10 **(d)** 9

9. The next palindrome after 3002 is: 1

(a) 3013 **(b)** 3003 **(c)** 2992 **(d)** 3113

10. The sum of the first 60 counting numbers $1 + 2 + 3 + \cdots + 60$ is: 1

(a) 3660 **(b)** 3600 **(c)** 1830 **(d)** 1890

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Apply the reverse-and-add steps on 96 and state the palindrome reached. 2

12. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 5, 1, 2, 3. Find the top number. 2

13. Find the greatest and the smallest 5-digit numbers formed from 9, 3, 7, 5, 1, each digit once. 2

14. Apply the reverse-and-add steps on 175 and state the palindrome reached. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Perform two Kaprekar steps on 4172. 3

16. Find the largest palindrome less than 3820 and the smallest palindrome greater than 3820. 3

17. Find the sum of all numbers from 1 to 80 by pairing the first and last numbers. 3

OR

Perform two Kaprekar steps on 5039. 3

18. Find the largest palindrome less than 5090 and the smallest palindrome greater than 5090. 3

OR

Find the sum of all numbers from 1 to 120 by pairing the first and last numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A 4-digit palindrome has first digit 4 and digit sum 10. Find the number. Add 11 to it and see if the answer is a palindrome. 5

OR

Using the digits 9, 1, 4, 6 once each, form a 4-digit number closest to 4500. Show the options you tried and the difference. 5

20. A 4-digit palindrome has first digit 5 and digit sum 22. Find the number. Add 11 to it and see if the answer is a palindrome. 5

OR

Using the digits 7, 3, 8, 2 once each, form a 4-digit number closest to 3000. Show the options you tried and the difference. 5

END OF QUESTION PAPER — Total: 40 Marks

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